

Hints & Solutions

#MathBoleTohMathonGo

Q1 (2) - Single Correct

Given parabolas are symmetric about the line $y = x$. So, they have a common normal with slope -1, it meets the parabolas at $\left(-\frac{1}{2}, \frac{13}{4}\right), \left(\frac{13}{4}, -\frac{1}{2}\right)$. Hence, the required circle, is

$$x^2 + y^2 - \frac{11}{4}x - \frac{11}{4}y - \frac{13}{4} = 0$$

Q2 (3) - Single Correct

Here, $y = ax^2 + bx + c$ has vertex (4,2)

$$\Rightarrow b = -8a$$

$$\text{and } \frac{4ac - b^2}{4a} = 2$$

$$\Rightarrow c = 2 + 16a$$

$$\therefore \text{product of } abc = \Delta = -16(a^2 + 8a^3)$$

$$\frac{d\Delta}{da} = -16(2a + 24a^2) < 0 \forall a \in [1, 3]$$

$$\Rightarrow \Delta_{\max} = -144, \Delta_{\min} = -3600$$

$$\therefore \text{difference} = 3456.$$

Q3 (1) - Single Correct

Line PQ is $y(t_1 + t_2) = 2x + 2at_1t_2$... (i)

and $\lambda y = -x + b$... (ii)

Since Eqs. (i) and (ii) are same

$$\Rightarrow \frac{t_1 + t_2}{\lambda} = \frac{2}{-1} = \frac{2at_1t_2}{b}$$

$$\therefore b = -at_1t_2 \text{ and } 2a \leq -at_1t_2 \leq 4a$$

$$\therefore -4 \leq t_1t_2 \leq -2$$

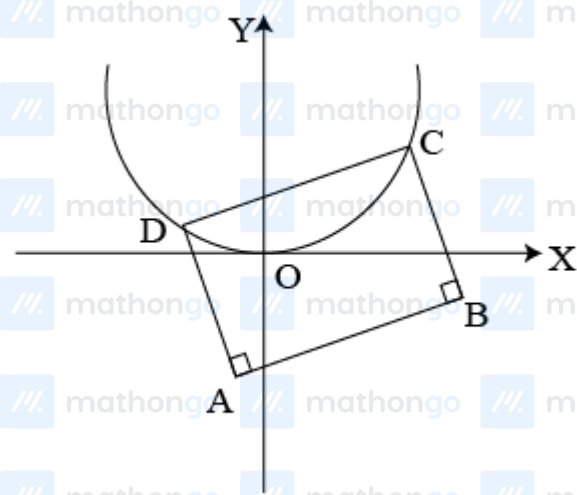
Q4 (2) - Single Correct

Let, $AB : y = 2x - 17$

$CD \Rightarrow y = 2x + C$

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where $C(x_1, y_1)$ and $D(x_2, y_2)$

$$m_{CD} = 2$$

$$\Rightarrow \frac{y_2 - y_1}{x_2 - x_1} = 2 \dots (i)$$

$$\text{Let, } CD = a = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

$$\Rightarrow a^2 = 5(x_1 - x_2)^2 \text{ from (i)}$$

$$= 5[(x_1 + x_2)^2 - 4x_1x_2] \dots (ii)$$

$$\Rightarrow x^2 - y = 2x + c$$

$$\Rightarrow x_1 + x_2 = 2$$

$$\text{and } x_1x_2 = -c \dots (iii)$$

from Eqs. (ii) & (iii)

$$a^2 = 5[4 + 4c] \dots (iv)$$

Let, $A = (k, 2k - 17)$ by any point on $y = 2x - 17$

$$\therefore AD = a = \frac{|2k - 2k + 17 + c|}{\sqrt{5}}$$

$$\Rightarrow a^2 = \frac{(17+c)^2}{5} \dots (v)$$

from (iv) & (v)

$$\Rightarrow a^2 = 1280$$

Q5 (1) - Single Correct

Normal is $y = mx - 2am + am^3$

$$\Rightarrow 0 = 2km - 2m - m^3$$

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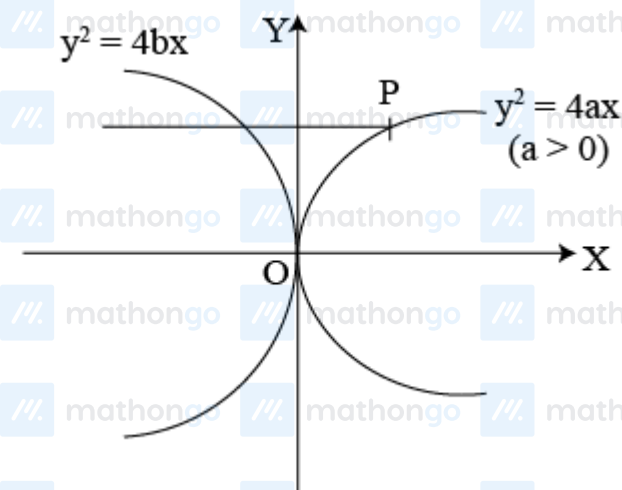
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Let m_1, m_2, m_3 be the roots Then $m_3 = 0$

$$\begin{aligned} \text{Then } m_3 &= 0 \\ \text{and } m_1 m_2 &= -1 \\ \Rightarrow k &= 3/2 \end{aligned}$$

Q6 (1) - Single Correct

$$P(at_1^2, 2at_1), P'(-bt_2^2, -2bt_2)$$



$$\text{Slope of } PP' = 0$$

$$\Rightarrow at_1 + bt_2 = 0 \dots (i)$$

$$\text{Mid-point of } PP' = \left(\frac{at_1^2 - bt_2^2}{2}, at_1 - bt_2 \right) = (h, k)$$

$$\begin{aligned} \text{put } t_2 &= -\frac{at_1}{b} \\ \Rightarrow \frac{k}{a} &= \frac{2\left(at_1 + \frac{at_1}{b}\right)}{at_1^2 - b\left(-\frac{at_1}{b}\right)^2} = \frac{4at_1}{a(b-a)t_1^2} \\ \therefore t_1 &= \frac{4h}{k(b-a)} \end{aligned}$$

$$\therefore k = at_1 - bt_2 = at_1 + rt_1 = 2a \frac{4h}{k(b-a)}$$

$$\Rightarrow k^2 = \frac{8a}{b-a} h$$

$$\therefore \text{locus of } (h, k) \text{ is } y^2 = \frac{8a}{b-a} \cdot x, \text{ a parabola}$$

Q7 (1) - Single Correct

$$(x - x_1)^2 + (y - y_1)^2 = r^2$$

$$\text{X-axis intercept} = 2a$$

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$$\Rightarrow 2\sqrt{x_1^2 - (x_1^2 + y_1^2 - r^2)} = 2a$$

$$\Rightarrow r^2 - y_1^2 = a^2$$

$$\therefore y_1^2 = r^2 - a^2$$

Passes, through $(0, b)$

$$\therefore x_1^2 + (b - y_1)^2 = r^2$$

$$\Rightarrow x_1^2 + b^2 - 2by_1 + y_1^2 - r^2 = 0$$

$$\Rightarrow x_1^2 + b^2 - 2by_1 - a^2 = 0$$

$\therefore$ locus of (x_1, y_1)

$$\Rightarrow x^2 - 2by + b^2 = x^2$$

Q8 (3) - Single Correct

Let the equation of circle be $x^2 + y^2 = a^2$ with ends of diameter $M(a, 0)$ and $M'(-a, 0)$

Equation of concentric circle is $x^2 + y^2 = b^2$ (i)

Equation of tangent to (i) is $y = mx + b\sqrt{m^2 + 1}$

$$mx - y + b\sqrt{m^2 + 1} = 0$$

which is the directrix of parabola Let the focus be $S(h, k)$

For parabola we have $PS = PM$

$$\sqrt{(h - a)^2 + k^2} = \frac{m(a) - 1(0) + b\sqrt{m^2 + 1}}{\sqrt{m^2 + 1}}$$

$$\sqrt{(h - a)^2 + k^2} = \frac{am + b\sqrt{m^2 + 1}}{\sqrt{m^2 + 1}} \dots\dots\dots \text{(ii)}$$

Also $PS = PM'$

$$\sqrt{(h + a)^2 + k^2} = \frac{m(-a) - 1(0) + b\sqrt{m^2 + 1}}{\sqrt{m^2 + 1}}$$

$$\sqrt{(h + a)^2 + k^2} = \frac{-am + b\sqrt{m^2 + 1}}{\sqrt{m^2 + 1}} \dots\dots\dots \text{(iii)}$$

Adding (ii) and (iii) $\sqrt{(h - a)^2 + k^2} + \sqrt{(h + a)^2 + k^2} = 2b$

$$\sqrt{(h + a)^2 + k^2} = 2b - \sqrt{(h - a)^2 + k^2}$$

squaring both sides

$$h^2 + 2ah + a^2 + k^2 = 4b^2 + h^2 - 2ah + a^2 + k^2 - 4b\sqrt{(h - a)^2 + k^2}$$

$$4b^2 - 4ah = 4b\sqrt{(h - a)^2 + k^2}$$

$$b^2 - ah = b\sqrt{(h - a)^2 + k^2}$$

$$b - \frac{ah}{b} = \sqrt{(h - a)^2 + k^2}$$

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Again squaring both sides $b^2 + \frac{a^2 h^2}{b^2} - 2ah = h^2 - 2ah + a^2 + k^2$

$$b^2 - a^2 = \left(1 - \frac{a^2}{b^2}\right) h^2 + k^2$$

$$\left(\frac{b^2 - a^2}{b^2}\right) h^2 + k^2 = b^2 - a^2$$

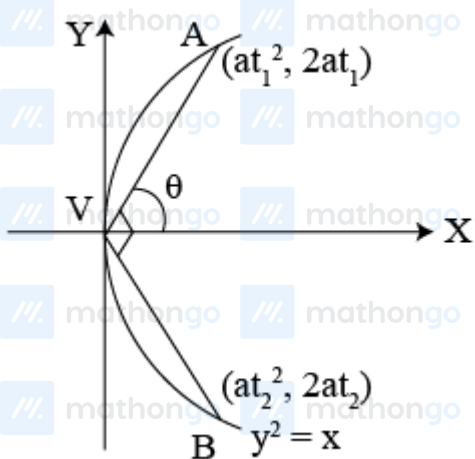
$$\frac{h^2}{b^2} + \frac{k^2}{b^2 - a^2} = 1$$

generalising the equation $\frac{x^2}{b^2} + \frac{y^2}{b^2 - a^2} = 1$

Q9 (4) - Single Correct

$$\tan \theta = \frac{2}{t_1} \dots (i)$$

$$\text{also, } \frac{2}{t_1} \cdot \frac{2}{t_2} = -1$$



$$\therefore t_1 t_2 = -4$$

$$|VA| = at_1 \sqrt{4 + t_1^2}$$

$$\text{using, } t_1 t_2 = -4$$

$$|VB| = \frac{4a}{t_1} \sqrt{\frac{16}{t_1^2} + 4} = \frac{8a}{t_1^2} \sqrt{4 + t_1^2}$$

$$\therefore \frac{|VA|}{|VB|} = \frac{at_1^3 \sqrt{4 + t_1^2}}{8a \sqrt{4 + t_1^2}} = \frac{t_1^3}{8}$$

$$\text{also, } \tan \theta = \frac{2}{t_1}$$

$$\Rightarrow t_1^3 = 8 \cot^3 \theta$$

$$\therefore \frac{|VA|}{|VB|} = \cot^3 \theta$$

Q10 (4) - Single Correct

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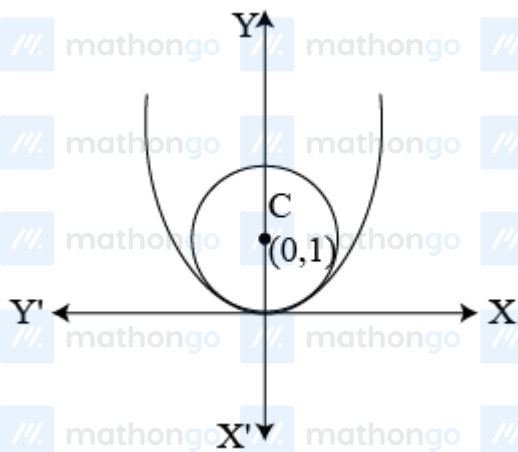
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Putting $x^2 = \frac{y}{a}$ in the circle $x^2 + (y-1)^2 = 1$, we get

$$\frac{y}{a} + y^2 - 2y = 0$$

(Note that for $a < 0$, they cannot intersect other than the origin)

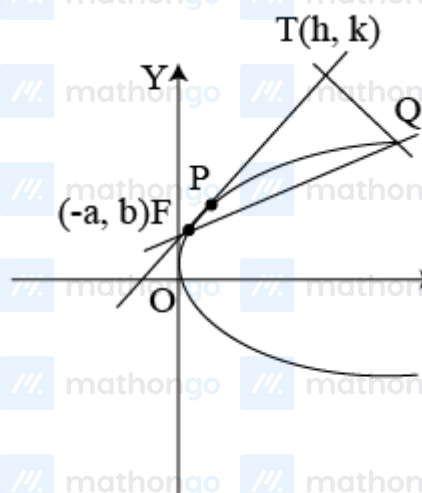
Hence, we get $y = 0$ or $y = 2 - \frac{1}{a}$



On substituting $y = 2 - \frac{1}{a}$ in $y = ax^2$, we get $ax^2 = 2 - \frac{1}{a} \Rightarrow x^2 = \frac{2a-1}{a^2} > 0 \Rightarrow a > \frac{1}{2}$

Q11 (3) - Single Correct

The chord of contact of (h, k) is $ky = 2a(x + h)$



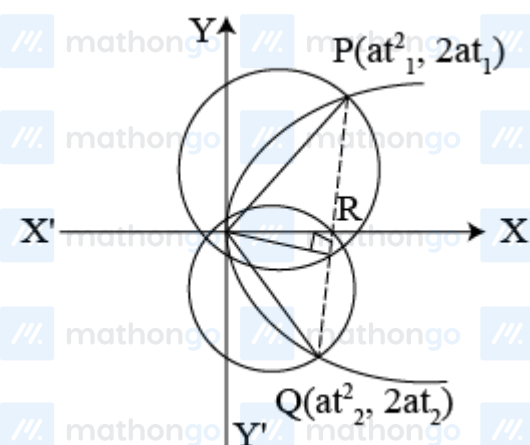
It passes through $(-a, b)$. Therefore, $bk = 2a(-a + h)$ Hence, the locus is $by = 2a(x - a)$

Q12 (1) - Single Correct

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The slope of tangent at P is $1/t_1$ and that at Q is $1/t_2$. Therefore, $\cot \theta_1 = t_1$ and $\cot \theta_2 = t_2$



$$\text{Slope of } PQ = \frac{2}{t_1 + t_2}$$

$$\therefore \text{Slope of } OR = -\left(\frac{t_1 - t_2}{2}\right) = \tan \phi \quad [\text{angle in semicircle is } 90^\circ]$$

$$\text{or } \tan \phi = -\frac{1}{2}[\cot \theta_1 + \cot \theta_2]$$

$$\therefore \cot \theta_1 + \cot \theta_2 = -2 \tan \phi$$

Q13 (1,2,3,4) - Multiple Correct

Any point on the parabola is $P(2t^2, 4t)$

Therefore, the mid-point of $S(2, 0)$ and $P(2t^2, 4t)$ is

$$R\left(\frac{2+2t^2}{2}, 2t\right) \equiv (h, k)$$

$$\therefore h = \frac{2+2t^2}{2}, k = 2t$$

$$\text{Eliminate } t \text{ i.e. } 2x = 2\left(1 + \frac{y^2}{4}\right) = 2 + \frac{y^2}{2}$$

$$\text{i.e. } 4x = 4 + y^2$$

$$\text{i.e. } y^2 = 4(x - 1)$$

It is a parabola with vertex at $(1, 0)$ and latusrectum 2.

$$\text{The directrix is } x - 1 = -1, \Rightarrow x = 0$$

$$\text{The focus is } x - 1 = 1, \Rightarrow x = 2$$

$$\text{i.e. its coordinates } (2, 0)$$

Q14 (1,4) - Multiple Correct

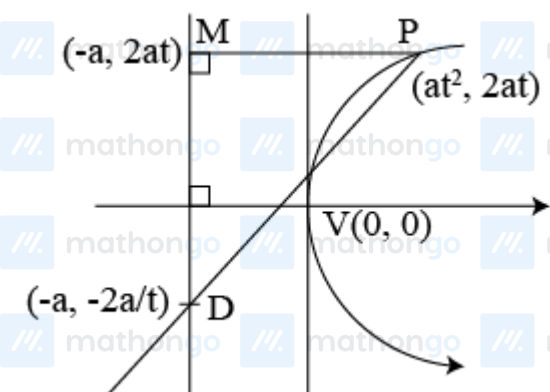
Here, Equation of VP is

$$\frac{y-0}{x-0} = \frac{2at}{at^2}$$

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$$\Rightarrow y = \frac{2}{t}x$$



which meets directrix at $x = -a$

$$\therefore y = \frac{2a}{t}$$

$$D\left(-a, \frac{-2a}{t}\right)$$

Equation of circle having DM as diameter is

$$(x+a)(x+a) + \left(y + \frac{2a}{t}\right)(y - 2at) = 0$$

For point of intersection used with X -axis, put $y = 0$

$$\therefore (x+a)^2 - 4a^2 = 0$$

$$x+a = \pm 2a \Rightarrow x = a, -3a$$

Q15 (2,3,4) - Multiple Correct

Here, $y = \tan(\tan^{-1}x) = x, x \in R$

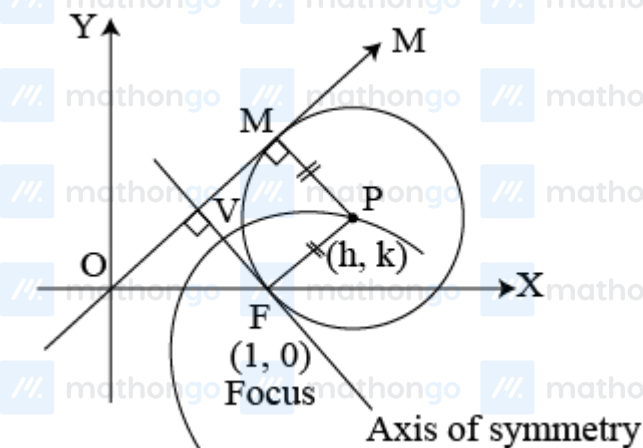
Since, $PF = PM$

$\therefore$ locus of P is parabola

Having, focus as $(1,0)$ and directrix as $x - y = 0$

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Distance between focus and directrix = $2a$

$$\Rightarrow 2a = \frac{|1-0|}{\sqrt{2}}$$

$$\text{Length of latusrectum} = 4a = \frac{2}{\sqrt{2}} = \sqrt{2}$$

Axis of symmetry is,

$$\frac{y-0}{x-1} = -1$$

$$\Rightarrow y = -x + 1$$

$$\text{or } x + y = 1$$

$$\text{Vertex} \left(1 + \frac{1}{2\sqrt{2}} \cos 135^\circ, 0 + \frac{1}{2\sqrt{2}} \sin 135^\circ \right) = \left(\frac{3}{4}, \frac{1}{4} \right)$$

Q16 (3,4) - Multiple Correct

Let $A(t_1^2, 2t_1)$ and $B(t_2^2, 2t_2)$ be two points on $y^2 = 4x$. The coordinates of the centre of the circle are $\left(\frac{t_1^2 + t_2^2}{2}, t_1 + t_2 \right)$

Since, the circle with diameter AB touches the axis of the parabola and is of radius r .

$$\therefore t_1 + t_2 = \pm r$$

$$\therefore \text{Slope of } AB = \frac{2}{t_2 + t_1} = \pm \frac{2}{r}$$

Q17 (2,3) - Multiple Correct

The equation of tangent to the parabola $y^2 = 4ax$ at $(at^2, 2at)$ is

$$ty = x + at^2$$

Here, $a = 3$. So, the equation of the tangent at $(3t^2, 6t)$ to the parabola $y^2 = 12x$ is

$$ty = x + 3t^2$$

The slope of Eq. (i) is $\frac{1}{t}$ and that of the given line is 3. Therefore,

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$$\frac{\frac{1}{t}-3}{1+\frac{1}{t} \times 3} = \pm \tan 45^\circ = \pm 1$$

$$\Rightarrow \frac{1-3t}{t+3} = \pm 1$$

$$\Rightarrow t = -\frac{1}{2}, 2$$

Now, equation of the tangents are

$$x - 2y + 12 = 0$$

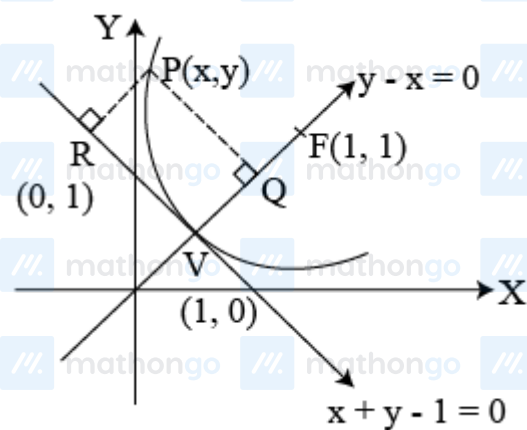
$$\text{and } 4x + 2y + 3 = 0$$

Q18 (2,3,4) - Multiple Correct

Here, axis of parabola is $y = x$

$$\text{Here, } VF = \frac{|1+1-1|}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\therefore \text{length of latusrectum} = \frac{4}{\sqrt{2}}$$



$\Rightarrow$ Equation of parabola, is

$$(PQ)^2 = 4a(PR)$$

$$\Rightarrow \left(\frac{x-y}{\sqrt{2}}\right)^2 = \frac{4}{\sqrt{2}} \left(\frac{x+y-1}{\sqrt{2}}\right)$$

$$\text{Or } (x-y)^2 = 4(x+y-1)$$

$$\Rightarrow \text{Vertex } \left(\frac{1}{2}, \frac{1}{2}\right), \text{ and length of latusrectum} = 2\sqrt{2}.$$

Q19 (1,4) - Multiple Correct

$$\text{As, } P(t^2, 2t)$$

$$\text{and } t^2 = 2t$$

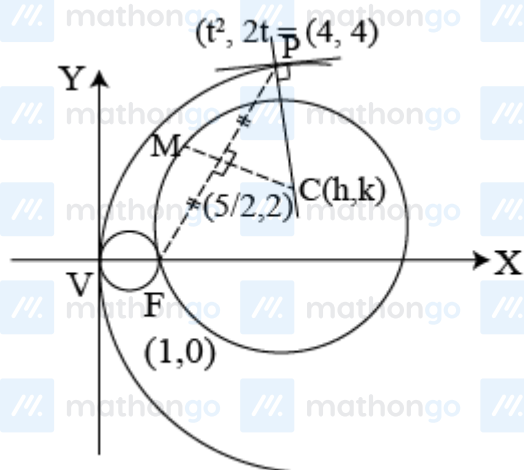
$$\Rightarrow t = 2$$

$$\text{Equation of Normal at } P \text{ is } y + xt = 2at + at^3$$

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$$\Rightarrow y + 2x = 4 + 8$$

$$\Rightarrow 2x + y = 12 \text{ passes through } (h, k)$$

$$\Rightarrow 2h + k = 12 \dots(i)$$

$$\text{and } m_{Ci} \cdot m_{PF} = -1$$

$$\Rightarrow \frac{k-2}{h-\frac{5}{2}} \cdot \frac{4-0}{4-1} = -1$$

$$\Rightarrow 4k - 8 = -3 \left(h - \frac{5}{2} \right)$$

$$\Rightarrow 3h + 4k = \frac{31}{2} \dots(ii)$$

On solving Eqs. (i) and (ii), we get $h = \frac{13}{2}, k = -1$

$\therefore$ Equation of circle, is

$$x^2 + y^2 - 13x + 2y + 12 = 0$$

$$\text{and } x^2 + y^2 - x = 0$$

when VF is diameter.

Q20 (1,2) - Multiple Correct

The equations of tangent to $y = x^2$ is

$$x = my + \frac{1}{4m} \dots(i)$$

The equation of tangent to $y = -(x-2)^2$ is

$$x - 2 = my - \frac{1}{4m} \dots(ii)$$

Lines given in Eqs. (i) and (ii) represent the same line.

$$\therefore \frac{1}{4m} = 2 - \frac{1}{4m}$$

$$\Rightarrow m = \frac{1}{4}$$

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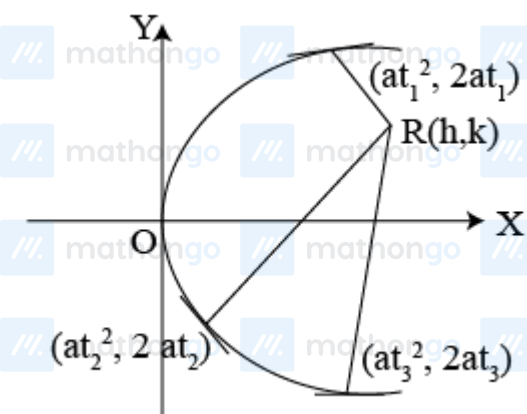
On substituting $m = \frac{1}{4}$ in Eq. (i), we get $y = 4(x - 1)$ as a common tangent.

Clearly, $y = x^2$ and $y = -(x - 2)^2$ represent parabolas having vertex at $(0, 0)$ respectively and touching X -axis. So, X -axis, i.e. $y = 0$ is also a common tangent.

Q21 (2) - Multiple Correct

Equation of normal is

$$y + xt = 2at + at^3 \dots (i)$$



as (i) passes through (h, k) , $(at_2^2, 2at_2)$

So, $at^3 + t(2a - h) - k = 0$ has roots $t_1, t_2, t_3 \dots (ii)$

Here, $a = 1$

$$\Rightarrow t_1 + t_2 + t_3 = 0 \dots (iii)$$

$$\text{also } \frac{2}{t_1 + t_2} = 2 \Rightarrow t_1 + t_2 = 1 \dots (iv)$$

from Eqs. (iii) & (iv)

$$t_3 = -1$$

Put, $t_3 = -1$ in Eq. (ii), we get

$$-1 - 1(2 - h) - k = 0$$

$\Rightarrow$ locus of $p(h, k)$ is $x - y = 3$.

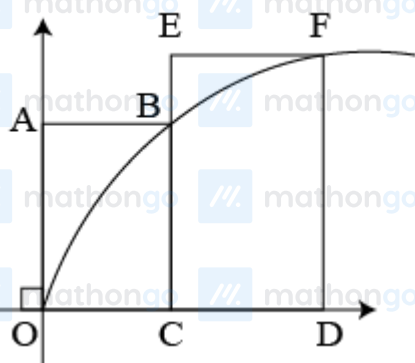
Q22 (1) - Multiple Correct

$$y^2 = k^2 x$$

$$\Rightarrow y^2 = 4ax, k^2 = 4a$$

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$$B(at_1^2, 2at_1), F(a_2^2, 2t_2)$$

$$(t_1, t_2 > 0)$$

$$\text{to find } \frac{FG}{BC} = \frac{2at_2}{2nt_1} = \frac{t_2}{t_1}$$

$$\text{now, } DC = BC$$

$$\Rightarrow at_1^2 = 2at \Rightarrow t_1 = 2$$

$$\text{also, } at_2^2 - at_1^2 = 2at_2$$

$$\Rightarrow t_2^2 - 4 = 2t_2$$

$$\text{or } t_2^2 - 2t_2 - 4 = 0$$

$$t_2 = \frac{2 \pm \sqrt{4+16}}{2}, \text{ neglecting } t_2 = \frac{2-\sqrt{20}}{2} \text{ as } t_2 > 0$$

$$\therefore t_2 = \frac{2+\sqrt{20}}{2}$$

$$\Rightarrow \frac{t_2}{t_1} = \frac{\sqrt{5}+1}{2}$$

Q23 (3) - Paragraph 1

Any point on the given parabola is $(t^2, 2t)$. The equation of tangent at $(1,2)$ [point at the upper end of the latusrectum] is

$$x - y + 1 = 0$$

The image (h, k) of the point $(t^2, 2t)$ on $x - y + 1 = 0$ is given by

$$\frac{n - t^2}{1} = \frac{k - 2t}{-1} = \frac{-2(t^2 - 2t + 1)}{1 + t}$$

$$\therefore h = t^2 - t^2 + 2t - 1 = 2t - 1$$

$$\text{and } k = 2t^2 + t^2 - 2t + 1 = t^2 + 1$$

$$\text{Eliminating } t \text{ from, } h = 2t - 1 \text{ and } k = t^2 + 1$$

$$\text{we get } (h+1)^2 = 4(k-1)$$

The required equation of reflection is

$$(x+1)^2 = 4(y-1)$$

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Q24 (2) - Paragraph 1

The image of $(t^2, 2t)$ in the tangent line and then eliminate t to get the image now family of circle touching the parabola

$$y^2 = 4x \text{ at } (1, 2)$$

$$(x - 1)^2 + (y - 2)^2 + \lambda(x - y + 1) = 0$$

it passes through $(1, 0)$

$$\Rightarrow 4 + 2\lambda = 0$$

$$\Rightarrow \lambda = -2$$

Hence circle is

$$x^2 + y^2 - 4x - 2y + 3 = 0$$

$$\therefore r = \sqrt{2}$$

Q25 (2) - Paragraph 2

Since, no point of the parabola is below the X -axis,

$$D = a^2 - 4 \leq 0$$

Therefore, the maximum value of a is 2. The equation of parabola, when $a = 2$ is $y = x^2 + 2x + 1$

It intersects the Y -axis at $(0, 1)$. The equation of tangent at $(0, 1)$ is

$$y = 2x + 1$$

Since, $y = 2x + 1$ touches the circle $x^2 + y^2 = r^2$, we get

$$r = \frac{1}{\sqrt{5}}$$

Q26 (1) - Paragraph 2

The equation of tangent at $(0, 1)$ to the parabola $y = x^2 + ax + 1$ is

$$y - 1 = a(x - 0)$$

$$\Rightarrow ax - y + 1 = 0$$

As it touches the circle, we get

$$r = \frac{1}{\sqrt{a^2 + 1}}$$

The radius is maximum, when $a = 0$. Therefore, the equation of the tangent is $y = 1$. Therefore, the slope of tangent is 0.

Q27 (1) - Paragraph 3

Area enclosed by tangent lines and parabola is equivalent to the area below the parabola from $[a, b]$ and above the line

$y = -1$ minus the area of two Δ 's first with vertices $(t, -1), (a, a^2), (a, -1)$ and the second with vertices $(t, -1), (b, b^2)$

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and $(b, -1)$

Sum of area of two triangles

$$= \frac{(a^2+1)(t-a)}{2} + \frac{(b^2+1)(b-t)}{2}$$

$$= 2(1+t^2)^{3/2}$$

$$S(t) = \int_a^b (x^2 + 1) dx - 2(1+t^2)^{3/2} = \frac{b^3 - a^3}{3} + (b-a) - 2(1+t^2)^{3/2}$$

$$= \frac{8}{3}(1+t^2)^{3/2} - 2(1+t^2)^{3/2} = \frac{2}{3}(1+t^2)^{3/2}$$

Q28 (1) - Paragraph 3

$$\frac{s(t)}{\sqrt{t}} \text{ is minimum } \Rightarrow \frac{d}{dt} \left(\frac{s(t)}{\sqrt{t}} \right) = 0$$

$$\Rightarrow t = \frac{1}{\sqrt{5}}$$

Q29 (3) - Paragraph 4

$$C_1 : y = (x-1)^2 + |k^2 - 3k + 2| + \frac{1}{4}$$

$$C_2 : (x-1)^2 = -4(y-2)$$

$$\text{for no point of intersection } |k^2 - 3k + 2| + \frac{1}{4} > 2$$

$$\Rightarrow |k^2 - 3k + 2| > \frac{7}{4}$$

Q30 (4) - Paragraph 4

$$f(k) \text{ is least when } k = \frac{3}{2}$$

$$\text{Least value of } f(k) = \frac{1}{4}$$

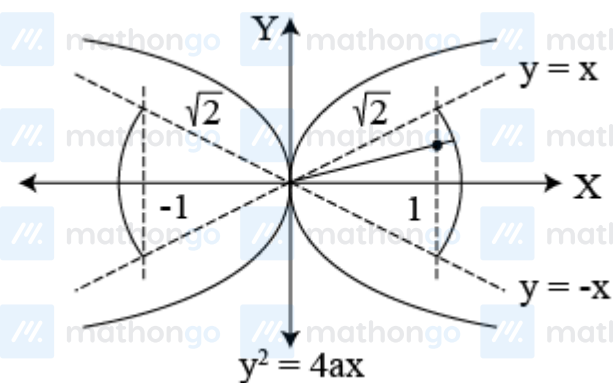
$$\text{foci } s_1 \left(1, \frac{1}{2}\right) \text{ and } s_2(1, 1) \Rightarrow s_1 s_2 = \frac{1}{2}$$

Q31 (2) - Paragraph 5

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$$y^2 = 4|x|$$

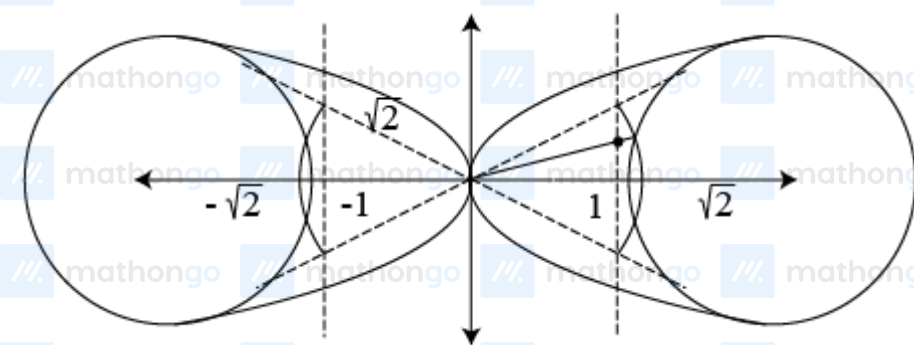


Here $y^2 = 4x$, directrix, $x = -1$

$\therefore$ point should lie for $y^2 = -4x$

$$\Rightarrow 1 < r < \sqrt{2}$$

Q32 (1) - Paragraph 5



$(x - a)^2 + y^2 = R^2$ touches $x^2 + y^2 = 1$ externally.

$a = 1 + y \Rightarrow y^2 = 4x$ and $(x - a)^2 + 4x - R^2 = 0$ has unique root.

$$\Rightarrow R^2 = 4(a - 1)$$

$$\text{or } R^2 = 4R \Rightarrow R = 4$$

Q33 (1) - Integer Type

Here, $A(at_1t_2, a(t_1 + t_2))$

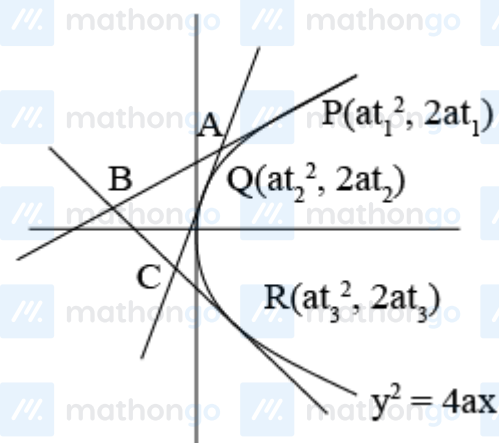
$B(at_1t_3, a(t_1 + t_3))$

$C(at_2t_3, a(t_2 + t_3))$

The equation of altitudes through the vertex A , is

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$$y - a(t_1 + t_2) = -t_3(x - at_1t_2)$$

$$\text{Or } y + xt_3 = a(t_1 + t_2 + t_1t_2t_3) \dots (i)$$

Similarly, the equation of altitudes from B and C , are

$$y + xt_2 = a(t_3 + t_1 + t_1t_2t_3) \dots (ii)$$

$$\text{and } y + xt_1 = a(t_2 + t_3 + t_1t_2t_3) \dots (iii)$$

Solving Eqs. (i) and (ii), we get

$$x = -a \text{ and } y = a(t_1 + t_2 + t_3 + t_1t_2t_3)$$

$$\therefore x = \lambda a \Rightarrow \lambda = -1 \text{ (as, } a = 1)$$

$$\text{or } |\lambda| = 1$$

Q34 (2) - Integer Type

The equation of the tangent of slope m_1 to $y^2 = 4a(x + a)$ is

$$y = m_1(x + a) + \frac{a}{m_1} \dots (i)$$

The equation of tangent of slope m_2 to

$$y^2 = 4b(x + b) \text{ is}$$

$$y = m_2(x + b) + \frac{b}{m_2} \dots (ii)$$

Since, Eqs. (i) and (ii) are perpendicular

$$\Rightarrow m_2 = -\frac{1}{m_1}$$

Putting $m_2 = -\frac{1}{m_1}$ in Eq. (i), we get

$$y = -\frac{1}{m_1}(x + b) - bm_1 \dots (iii)$$

Subtracting Eqs. (iii) and (i), we get

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$$0 = \left(m_1 + \frac{1}{m_1}\right)x + a\left(m_1 + \frac{1}{m_1}\right) + b\left(m_1 + \frac{1}{m_1}\right)$$

$$0 = (x + a + b)\left(m_1 + \frac{1}{m_1}\right)$$

$$x + a + b = 0$$

$$\therefore \lambda = \mu = 1$$

$$\Rightarrow \lambda + \mu = 2$$

Q35 (6) - Integer Type

Parabolas are equal, if their latusrectum are equal.

Latusrectum of $y^2 = \lambda x$ is $\lambda(x - 3)^2 + (y - (-2))^2$

$$= \left(\frac{3x - 4y - 2}{5}\right)^2$$

$\therefore$ focus (3,-2)

directrix, $3x - 4y - 2 = 0$

$$3z = 2a = 3 \Rightarrow \frac{\lambda}{2} = 3 \Rightarrow \lambda = 6$$

Q36 (3) - Integer Type

$$\text{Let } P(t) \Rightarrow Q\left(-\frac{1}{4}\right)$$

$$P' = -t - \frac{2}{t}, Q' = \frac{1}{t} + 2t$$

$$\Rightarrow P'Q' = a \left| 3 \left(t + \frac{1}{t}\right) \right| \sqrt{\left(t - \frac{1}{t}\right)^2 + 4} = 3a \left(t + \frac{1}{t}\right)^2$$

$$\text{and } PQ = a \left(t + \frac{1}{t}\right)^2$$

$$\Rightarrow \frac{P'Q'}{PQ} = 3$$

Q37 (8) - Integer Type

The curve is $(y + 1)^2 = 4(x - 1)$

Equation of the normal to the given curve is

$$y + 1 = m(x - 1) - 2m - m^3$$

which passes through (h, k)

$$\Rightarrow m^3 + m(3 - h) + (1 + k) = 0$$

$$m_1 m_2 m_3 = -(1 + k) \Rightarrow m_3 = 1 + k [\because m_1 m_2 = -1]$$

$$\Rightarrow (k + 1)^3 + (1 + k)(3 - h) + (1 + k) = 0$$

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$$\Rightarrow (k+1)^3 + 3 - h + 1 = 0$$

$$\text{Or } (y+1)^2 = x - 4$$

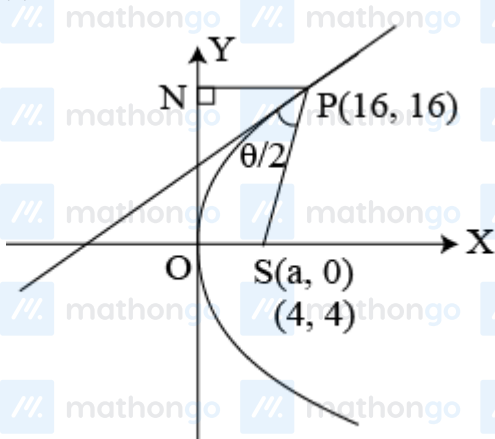
$$\therefore \frac{\mu}{2} = -1, \mu = -2, \lambda = 4$$

$$\therefore \lambda - 2\mu = 8$$

Q38 (2) - Integer Type

Since PT bisects $\angle NPS$.

$$\therefore \angle NPS = \angle PSX$$



$$\Rightarrow \tan^{-1}\left(\frac{16}{12}\right) = \tan^{-1}\left(\frac{4}{3}\right) = \theta$$

$$\Rightarrow \tan \theta = \frac{4}{3} = \frac{2 \tan \theta/2}{1 - \tan^2 \theta/2}$$

$$\Rightarrow \tan \frac{\theta}{2} = \frac{1}{2}$$

$$\therefore \angle TPS = \cot^{-1}(2)$$

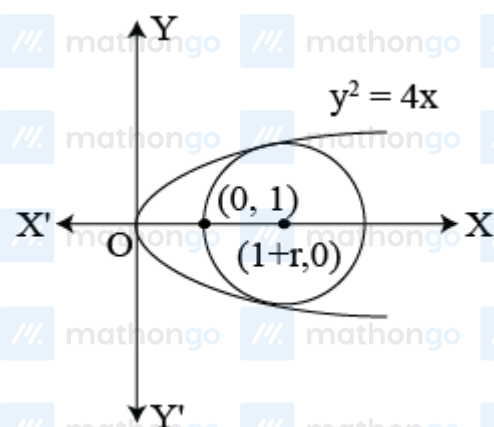
$$\therefore k = 2$$

Q39 (4) - Integer Type

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Let r be the radius of the largest circle passing through the focus $(1,0)$ of $y^2 = 4x$



Clearly, centre of the circle will be on X -axis and its coordinates are $(1+r, 0)$

The equation of circle is $(x - 1 - r)^2 + y^2 = r^2$.

It touches $y^2 = 4x$. Therefore, the equation $(x - r - 1)^2 + 4x = r^2$ must have equal roots. $\Rightarrow 4(1-r)^2 - 4(2r+1) = 0$

$$\therefore r = 4$$

Q40 (2) - Integer Type

Let $P(h, k)$, $(k \neq 0)$ through which normals are drawn to the parabola

$$y^2 = 4ax \quad (a = 1)$$

such that product of slopes of two is α

$$m_1 m_2 = \alpha$$

Now, any normal to the parabola is

$$y = mx + 2am - am^3$$

$$\Rightarrow am^3 + 0m^2 + m(2a - h) + k = 0$$

$$\Rightarrow m^3 + 0m^2 + m(2a - h) + k = 0, \text{ as } a = 1 \dots (i)$$

$$m_1 m_2 m_3 = -k$$

$$\therefore m_3 = \frac{-k}{m_1 m_2} = \frac{-k}{\alpha} \text{ [by Eqs. (i)]}$$

But m_3 is a root of Eq. (i).

$$\therefore k^3 + (2 - h)k\alpha^2 - k\alpha^3 = 0$$

Hence, the locus of P is

$$y^2 + (2 - x)\alpha^2 - \alpha^3 = 0 \quad [k \neq 0, \text{ cancelled}]$$

Above is same as $y^2 - 4x = 0$, as given.

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Hence, on comparing $\alpha^2 = 4$ and $2\alpha^2 - \alpha^3 = 0$

and $\alpha = 2$

$\therefore \alpha = 2$, which satisfies both.

Q41 (5) - Integer Type

$$m = \left(\frac{dy}{dx} \right)_{(1,1)} = b + 2$$

Equation of the tangent, is

$$(y - 1) = (b + 2)(x - 1)$$

$$\Rightarrow (b + 2)x - y - (b + 1) = 0$$

$$\Rightarrow \frac{(b+1)^2}{2|b+2|} = 2$$

$$\Rightarrow b^2 + 2b + 1 = \pm 4(b + 2)$$

$$\Rightarrow b^2 - 2b - 7 = 0$$

$$\text{and } b^2 - 6b + 9 = 0$$

$$\Rightarrow b_1 = -3$$

$$b_2 = 1 - \sqrt{2}$$

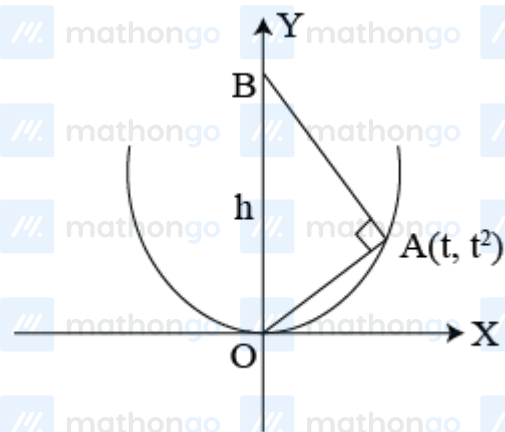
$$b_3 = 1 + \sqrt{2}$$

$$\Rightarrow |b_1| + |b_2 + b_3| = 5$$

Q42 (1) - Integer Type

Slope of $OA = t$

$$\Rightarrow \text{Slope of } AB = -\frac{1}{t}$$



Equation of AB ,

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$$y - t^2 = -\frac{1}{t}(x - t^2)$$

$$\text{put, } x = 0$$

$$\Rightarrow h = t^2 + 1$$

$$\text{as } x \rightarrow 0, t \rightarrow 0$$

$$\Rightarrow h \rightarrow 1$$

Q43 (8) - Integer Type

Given, equation of parabola is $y^2 = 8x$.

Put $y = \frac{4x+4}{3}$ in the equation of parabola.

$$\therefore \left(\frac{4x+4}{3}\right)^2 = 8x$$

$$\Rightarrow \frac{16(x+1)^2}{9} = 8x$$

$$\Rightarrow 2x^2 - 5x + 2 = 0$$

$$\Rightarrow x = 2, \frac{1}{2}$$

$$\text{when } x = 2, y = 4$$

$$\text{and } x = \frac{1}{2}, y = 2$$

Mid-point of (2,4) and $\left(\frac{1}{2}, 2\right)$ is $\left(\frac{5}{4}, 3\right)$.

On comparing with (λ, μ) , we get

$$\lambda = \frac{5}{4}, \mu = 3$$

$$4\lambda + \mu = 4 \times \frac{5}{4} + 3$$

$$= 8$$

Q44 (2) - Integer Type

Let (h, k) be a point then the polar of this point with respect to $y^2 = 4x$ axis

$$ky = 2(x + h) \text{ or } 2ax - ky + 2ah = 0 \dots(i)$$

This touches the parabola $x^2 = 4y$

So, it should be of the form

$$x = my + b/m \dots(ii)$$

Comparing (i) and (ii), we get

$$m = k/2a \text{ and } b/m = -2ah/2a$$

$$\Rightarrow -b/h = k/2a$$

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$$\Rightarrow hk = -2ab$$

Or $xy = -2ab$, where $ab = 1$

$$\Rightarrow \lambda = -2$$

$$\text{Or } |\lambda| = 2$$

Q45 (4) - Integer Type

The equation of any tangent to $y^2 = 4ax$ is

$$y = mx + \frac{a}{m} \dots (i)$$

Let (h, k) be the pole of Eq. (i) with respect to the parabola $y^2 = 4bx$. Then, the equation of polar is

$$ky = 2b(x + h)$$

$$\Rightarrow 2bx - ky + 2bh = 0 \dots (ii)$$

Clearly, Eqs. (i) and (iii) represent the same straight line. $\frac{2b}{m} = \frac{-k}{-l} = \frac{2bh}{a/m}$

$$\Rightarrow k = \frac{2b}{m}$$

$$\text{and } h = \frac{a}{m^2}$$

$$\Rightarrow m = \frac{2b}{k}$$

$$\text{and } m^2 = \frac{a}{h}$$

$$\Rightarrow \left(\frac{2b}{k}\right)^2 = \frac{a}{h}$$

$$\Rightarrow k^2 = \frac{4b^2h}{a}$$

Hence, the locus of (h, k) is $y^2 = \frac{4b^2x}{a}$.

But according to the question,

$$y^2 = \frac{\lambda b^2 x}{a}$$

$$\therefore \frac{4b^2 x}{a} = \frac{\lambda b^2 x}{a}$$

$$\Rightarrow \lambda = 4$$

Q46 (3) - Integer Type

The equation of any normal to $y^2 = 4ax$ is $y = mx - 2m - m^3$.

If it passes through $(2k, 0)$, then

$$m^3 + 2m(1 - k) = 0$$

$$\Rightarrow m = 0 \text{ or } m^2 + 2(1 - k) = 0$$

Clearly, $m = 0$ gives $y = 0$, i.e. X -axis as a normal. Other two normals will be perpendicular, if product of the roots of

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$$m^2 + 2(1 - k) = 0 \text{ is } -1$$

$$\text{i.e. } 2(1 - k) = -1 \Rightarrow k = \frac{3}{2}$$

$$\text{Hence, the value of } 2k \text{ is } 2 \times \frac{3}{2} = 3.$$

Q47 (2) - Integer Type

$$\text{Let } y = mx - 2am - am^3 \text{ passes } (h, k)$$

$$\Rightarrow k = mh - 2am - am^3$$

$$\Rightarrow am^3 + m(2a - h) + k = 0, \text{ having roots } m_1, m_2, m_3$$

$$\Rightarrow m_1 + m_2 + m_3 = 0$$

$$m_1 m_2 + m_2 m_3 + m_3 m_1 = \frac{2a - h}{a}$$

$$m_1 \cdot m_2 \cdot m_3 = -\frac{k}{a}$$

We have,

$$\tan^{-1} m_1 + \tan^{-1} m_2 + \tan^{-1} m_3 = \text{constant}(= \lambda \text{ say})$$

$$\Rightarrow \tan^{-1} \left[\frac{\Sigma m_1 - m_1 m_2 m_3}{1 - \Sigma m_1 m_2} \right] = \lambda$$

$$\Rightarrow \frac{k}{h - a} = \tan \lambda$$

$$\Rightarrow \mu = 2$$

Q48 (4) - Integer Type

$$\text{Let the normal at } P(at^2, 2at) \text{ meets the parabola again at } Q(at_1^2, 2at_1)$$

$$\text{Then, } t_1 = -t - \frac{2}{t}.$$

Now, distance of Q from the axis of the parabola

$$= |2at_1| = 2a |t_1|$$

$$= 2a \left| -t - \frac{2}{t} \right|$$

$$= 2a \left| t + \frac{2}{t} \right| \geq 2a \left| 2\sqrt{t \times \frac{2}{t}} \right| \geq 4a\sqrt{2} \quad [\because \text{AM} \geq \text{GM}]$$

$$\text{But according to the question, Distance of } Q \text{ from the axis of the parabola} = \lambda\sqrt{2a}$$

$$\text{But } 4a\sqrt{2} = \lambda\sqrt{2a}$$

$$\therefore \lambda = 4$$